Use Case Flow Document

Online Bookstore

1. Introduction

This document describes the use cases for the Online Bookstore system. It identifies the actors, provides a use case summary table, and details the flow of events (main flow, alternative flows, and exception flows) for each use case. The use cases are derived from the Functional Requirements (FR-1 to FR-15) in the Software Requirements Specification and are visualised in UseCaseDiagram.drawio / UseCaseDiagram.png in the same folder.

2. Actors

Actor	Description
Buyer	A registered or guest customer who browses, searches, and purchases books, tracks orders, and writes reviews.
Seller	A registered user who lists books for sale and manages inventory and incoming orders for their listings.
Admin	A platform moderator who approves, suspends, or removes listings and user accounts.
Payment Gateway	External system that authorises and captures payments and returns the transaction result.
Email / Notification Service	External system that delivers order confirmations, OTPs, and status notifications to users.

3. Use Case Summary Table

Use Case ID	Use Case Name	Primary Actor(s)	FR Reference
UC1	Register / Log In	Buyer, Seller, Admin	FR-1
UC2	Reset Forgotten Password	Buyer, Seller, Admin	FR-14
UC3	Search Books	Buyer	FR-2
UC4	View Book Details	Buyer	FR-3
UC5	Manage Shopping Cart	Buyer	FR-4
UC6	Manage Wishlist	Buyer	FR-15
UC7	Apply Discount Code	Buyer	FR-11
UC8	Place Order & Checkout	Buyer	FR-5, FR-6, FR-13
UC9	Process Payment	Buyer, Payment Gateway	FR-5
UC10	Generate Invoice / Receipt	System, Buyer	FR-13

Use Case ID	Use Case Name	Primary Actor(s)	FR Reference
UC11	Send Order Confirmation	System, Email / Notification Service	FR-6
UC12	Track Order Status	Buyer	FR-9
UC13	Rate & Review Purchased Book	Buyer	FR-10
UC14	Manage Book Listings	Seller	FR-7
UC15	Manage Incoming Orders	Seller	FR-8
UC16	Moderate Listings & Accounts	Admin	FR-12

4. Detailed Use Case Flows

UC1 - Register / Log In

Use Case ID	UC1
Use Case Name	Register / Log In
Primary Actor	Buyer, Seller, Admin
Secondary Actors	Email / Notification Service
Trigger	User selects Register or Log In.
Preconditions	The user has access to the application; for login, an account already exists.
Postconditions	A new account is created (register) or an authenticated session token is issued (login); user is redirected to the home page / dashboard.
Main Flow	1. User opens the Register / Log In page. 2. For registration: user enters email, phone, and password. 3. System validates format and password strength and checks the email/phone is not already registered. 4. System stores the account with a hashed password. 5. For login: user enters registered email/phone and password. 6. System verifies the credentials against stored values. 7. System generates a session token and redirects the user to the home page.
Alternative Flows	3a. Email/phone already registered: system shows "Account already exists" and returns to step 2. 6a. User has forgotten the password: user is routed to UC2 Reset Forgotten Password.
Exception Flows	E1. Invalid email format or weak password: system shows a validation error and blocks submission. E2. 3 consecutive failed logins: system shows a lockout warning and applies a temporary lockout per the security policy. E3. Mandatory field (e.g. phone) blank: system shows "<field> is required".

UC2 - Reset Forgotten Password

Use Case ID	UC2
Use Case Name	Reset Forgotten Password

Primary Actor	Buyer, Seller, Admin
Secondary Actors	Email / Notification Service
Trigger	User selects "Forgot password".
Preconditions	An account exists for the supplied email/phone.
Postconditions	The account password is replaced with the new value; old sessions may be invalidated.
Main Flow	1. User enters the registered email/phone. 2. System generates a one-time link/OTP and sends it via the Email / Notification Service. 3. User submits the OTP / opens the link. 4. System verifies the OTP is valid and not expired. 5. User enters and confirms a new password. 6. System validates strength and stores the new hashed password. 7. System confirms the reset and prompts the user to log in.
Alternative Flows	1a. Email/phone not found: system shows a generic "If the account exists, a reset link was sent" message (no account enumeration).
Exception Flows	E1. OTP expired or incorrect: system rejects it and offers to resend. E2. New password fails the strength policy: system shows the error and stays on step 5.

UC3 - Search Books

Use Case ID	UC3
Use Case Name	Search Books
Primary Actor	Buyer
Trigger	Buyer submits a search query or applies a filter.
Preconditions	The catalogue contains at least one book.
Postconditions	A result list matching the criteria is displayed.
Main Flow	1. Buyer enters a keyword (title, author, or ISBN) and/or selects genre and price range. 2. System queries the catalogue against the criteria. 3. System returns the matching books with key details and pagination.
Alternative Flows	1a. Empty query: system shows "Enter a search term" and does not run the query. 3a. No matches: system shows a "No results found" message with suggestions.
Exception Flows	E1. Catalogue service unavailable: system shows a retry message.

UC4 - View Book Details

Use Case ID	UC4
Use Case Name	View Book Details
Primary Actor	Buyer
Trigger	Buyer selects a book from search results or a listing.

Preconditions	The selected book exists and is visible (not removed/suspended).
Postconditions	The full book detail page is displayed.
Main Flow	1. Buyer selects a book. 2. System retrieves price, availability, seller, condition, and reviews. 3. System displays the book detail page with Add to Cart / Add to Wishlist actions.
Alternative Flows	2a. Book out of stock: system shows "Currently unavailable" and disables Add to Cart.
Exception Flows	E1. Book was removed by Admin/Seller: system shows a "Listing no longer available" page.

UC5 - Manage Shopping Cart

Use Case ID	UC5
Use Case Name	Manage Shopping Cart
Primary Actor	Buyer
Trigger	Buyer adds, updates, or removes a cart item.
Preconditions	Buyer is identified (guest or logged in).
Postconditions	The cart contents and totals reflect the change.
Main Flow	1. Buyer selects Add to Cart on a book, or opens the cart. 2. System adds the item / updates the quantity / removes the item. 3. System recalculates the subtotal and item count. 4. System displays the updated cart.
Alternative Flows	2a. Requested quantity exceeds available stock: system caps to the available quantity and notifies the buyer.
Exception Flows	E1. Item became unavailable while in the cart: system flags it and excludes it from checkout.

UC6 - Manage Wishlist

Use Case ID	UC6
Use Case Name	Manage Wishlist
Primary Actor	Buyer
Trigger	Buyer adds or removes a book from the wishlist.
Preconditions	Buyer is logged in.
Postconditions	The wishlist is updated and persisted to the buyer's account.
Main Flow	1. Buyer selects Add to Wishlist on a book, or opens the wishlist. 2. System adds or removes the book. 3. System persists and displays the updated wishlist.
Alternative Flows	1a. Buyer is not logged in: system routes to UC1 and resumes after login.
Exception Flows	E1. Book already on the wishlist: system ignores the duplicate.

UC7 - Apply Discount Code

Use Case ID	UC7
Use Case Name	Apply Discount Code
Primary Actor	Buyer
Trigger	Buyer enters a coupon/discount code at checkout.
Preconditions	Buyer is at the checkout step with at least one payable item.
Postconditions	A valid discount is applied and the order total is reduced.
Main Flow	1. Buyer enters a discount code. 2. System validates the code (exists, active, within usage limits, meets the cart conditions). 3. System applies the discount and recalculates the order total. 4. System shows the discount line and the new total.
Alternative Flows	2a. Code expired or invalid: system shows "Invalid or expired code" and leaves the total unchanged. 2b. Cart does not meet the minimum spend: system shows the condition and does not apply the code.
Exception Flows	E1. Code already used by this buyer: system rejects it as already redeemed.

UC8 - Place Order & Checkout

Use Case ID	UC8
Use Case Name	Place Order & Checkout
Primary Actor	Buyer
Secondary Actors	Payment Gateway, Email / Notification Service
Trigger	Buyer selects Checkout from the cart.
Preconditions	Buyer is logged in; the cart has at least one available item; a shipping address is set.
Postconditions	An order is created with status "Placed"; payment is captured; an invoice is generated; a confirmation is sent.
Main Flow	1. Buyer reviews the cart, shipping address, and delivery option. 2. Buyer optionally applies a discount code (UC7 - extend). 3. System calculates the final total (items + shipping - discount + tax). 4. System invokes UC9 Process Payment (include). 5. On payment success, system creates the order with status "Placed" and decrements stock. 6. System invokes UC10 Generate Invoice / Receipt (include). 7. System invokes UC11 Send Order Confirmation (include). 8. System shows the order confirmation page with the order number.
Alternative Flows	2a. No discount code: skip to step 3. 5a. Stock changed during checkout: system removes the affected line, recalculates, and asks the buyer to confirm before payment.

Exception Flows	E1. Payment declined/failed (from UC9): no order is created; system shows the failure reason and returns to the payment step. E2. Order created but confirmation delivery fails: order stands; system queues the notification for retry.

UC9 - Process Payment

Use Case ID	UC9
Use Case Name	Process Payment
Primary Actor	Buyer
Secondary Actors	Payment Gateway
Trigger	Included by UC8 after the buyer confirms the order total.
Preconditions	A pending order total and a selected payment method exist; the connection to the Payment Gateway is available.
Postconditions	The payment is authorised and captured, or the transaction is marked failed.
Main Flow	1. System sends the amount, currency, and payment details to the Payment Gateway over TLS. 2. Payment Gateway authorises and captures the payment. 3. Payment Gateway returns a success result with a transaction reference. 4. System records the transaction reference against the order.
Alternative Flows	2a. Gateway requires additional authentication (e.g. 3-D Secure): buyer completes the challenge and the flow resumes.
Exception Flows	E1. Payment declined: system returns a failure result to UC8. E2. Gateway timeout / no response: system marks the transaction "unknown", does not create the order, and advises the buyer to retry.

UC10 - Generate Invoice / Receipt

Use Case ID	UC10
Use Case Name	Generate Invoice / Receipt
Primary Actor	System
Secondary Actors	Buyer
Trigger	Included by UC8 after a successful payment.
Preconditions	An order exists with a captured payment.
Postconditions	An invoice/receipt is generated, stored, and made available to the buyer.
Main Flow	1. System compiles order lines, taxes, discounts, totals, and the transaction reference. 2. System generates the invoice document and assigns an invoice number. 3. System attaches the invoice to the order and makes it downloadable.
Alternative Flows	None.
Exception Flows	E1. Invoice generation fails: system logs the error and retries asynchronously; the order is unaffected.

UC11 - Send Order Confirmation

Use Case ID	UC11
Use Case Name	Send Order Confirmation
Primary Actor	System
Secondary Actors	Email / Notification Service
Trigger	Included by UC8 once the order is placed.
Preconditions	An order with status "Placed" exists and the buyer has a valid contact address.
Postconditions	A confirmation email/notification with the order summary is dispatched.
Main Flow	1. System composes the confirmation with the order number, items, total, and delivery estimate. 2. System hands the message to the Email / Notification Service. 3. Service accepts the message for delivery. 4. System marks the confirmation as sent.
Alternative Flows	None.
Exception Flows	E1. Service rejects or fails: system queues the message and retries with backoff.

UC12 - Track Order Status

Use Case ID	UC12
Use Case Name	Track Order Status
Primary Actor	Buyer
Trigger	Buyer opens an order from order history.
Preconditions	Buyer is logged in and owns at least one order.
Postconditions	The current order status and history are displayed.
Main Flow	1. Buyer selects an order. 2. System retrieves the status (Placed, Shipped, Out for delivery, Delivered) and timestamps. 3. System displays the status timeline and tracking reference if available.
Alternative Flows	2a. Order still "Placed": system shows only the placed step as complete.
Exception Flows	E1. Order not found / not owned by the buyer: system shows an authorization error.

UC13 - Rate & Review Purchased Book

Use Case ID	UC13
Use Case Name	Rate & Review Purchased Book
Primary Actor	Buyer
Trigger	Buyer submits a rating and review for a book.
Preconditions	Buyer is logged in and has a delivered order containing the book.
Postconditions	The review is stored and reflected in the book's aggregate rating.

Main Flow	1. Buyer opens a delivered book and selects Write a Review. 2. Buyer enters a star rating and review text. 3. System validates ownership (verified purchase) and content length. 4. System stores the review and recomputes the book's average rating. 5. System displays the published review.
Alternative Flows	3a. Buyer has already reviewed this book: system loads the existing review for editing instead of creating a new one.
Exception Flows	E1. No verified purchase: system blocks the review with an explanation. E2. Review flagged by content filter: system holds it for moderation.

UC14 - Manage Book Listings

Use Case ID	UC14
Use Case Name	Manage Book Listings
Primary Actor	Seller
Trigger	Seller adds, edits, or removes a listing.
Preconditions	Seller is logged in with an approved seller account.
Postconditions	The listing is created, updated, or removed and its visibility reflects moderation status.
Main Flow	1. Seller opens the listings dashboard. 2. Seller selects Add, Edit, or Remove. 3. Seller enters/updates title, author, ISBN, price, condition, and stock. 4. System validates the data and saves the listing. 5. System marks a new listing as "Pending approval" (see UC16) and shows it in the dashboard.
Alternative Flows	2a. Remove: system asks for confirmation, then hides the listing and cancels open reservations.
Exception Flows	E1. Required field missing or invalid ISBN: system shows a validation error and does not save. E2. Listing has open orders: system prevents deletion until those orders are resolved.

UC15 - Manage Incoming Orders

Use Case ID	UC15
Use Case Name	Manage Incoming Orders
Primary Actor	Seller
Secondary Actors	Email / Notification Service
Trigger	Seller opens the incoming orders queue or updates an order's fulfilment status.
Preconditions	Seller is logged in; at least one order exists for the seller's listings.
Postconditions	The order fulfilment status is updated and the buyer is notified.

Main Flow	1. Seller opens the incoming orders list. 2. Seller selects an order and reviews items and shipping details. 3. Seller updates the status (e.g. Accepted, Shipped) and adds a tracking reference. 4. System saves the status and triggers a buyer notification.
Alternative Flows	3a. Seller cannot fulfil: seller cancels the line; system initiates a refund and notifies the buyer.
Exception Flows	E1. Status transition not allowed (e.g. Delivered -> Accepted): system rejects the change.

UC16 - Moderate Listings & Accounts

Use Case ID	UC16
Use Case Name	Moderate Listings & Accounts
Primary Actor	Admin
Secondary Actors	Email / Notification Service
Trigger	Admin reviews a pending listing, a report, or a user account.
Preconditions	Admin is logged in with moderation privileges.
Postconditions	The listing or account is approved, suspended, or removed and the owner is notified.
Main Flow	1. Admin opens the moderation queue. 2. Admin selects a listing or account and reviews its details and any reports. 3. Admin chooses Approve, Suspend, or Remove and enters a reason. 4. System applies the decision and updates visibility/access. 5. System notifies the owner of the outcome and reason.
Alternative Flows	3a. Approve: listing becomes publicly visible; account regains full access. 3b. Suspend: item is hidden temporarily and can be reinstated later.
Exception Flows	E1. Target already actioned by another admin: system shows the current state and cancels the duplicate action.